CS23U02

Perspectives and Sustainability Development

ECOLOGICAL FOOTPRINT & ENVIRONMENTAL IMPACT

Ecological Footprint of Students' Lifestyle

Gryffindors

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1. INTRODUCTION

Climate change and environmental degradation demand immediate action. An ecological footprint measures how individual lifestyles impact natural resources and ecosystems.

Students represent a major population whose daily habits, transportation, energy usage, food choices, and plastic consumption, collectively create a significant environmental impact.

Gryfindors is an interactive web application designed to:

- Measure individual ecological footprints
- Provide actionable sustainability insights
- Enable institutions to analyze campus-wide environmental trends

2. PROBLEM STATEMENT

Despite awareness of sustainability, students lack:

- Practical tools to measure their environmental impact
- Clear understanding of how daily habits affect carbon emissions
- Immediate feedback to drive behavioral change

This results in:

- Low engagement in sustainable practices
- Lack of data-driven decision-making at the institutional level

3. OBJECTIVES

- Quantify Impact: Calculate ecological footprint based on lifestyle factors
- Personal Feedback: Classify users into Low, Moderate, High impact
- Actionable Tips: Provide personalized sustainability recommendations
- Campus Analytics: Analyze collective student data via CSV
- Policy Support: Enable data-driven sustainability initiatives

4. PROPOSED SOLUTION

A Streamlit-based web application with two core modules:

Individual Calculator

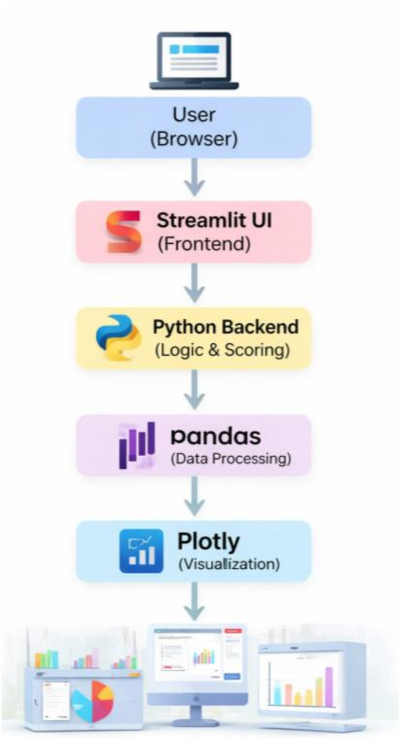
- User inputs lifestyle data

- Instant footprint score & impact level
- Visual breakdown (transport, energy, food, plastic)
- Personalized sustainability tips

College Data Analysis

- Upload Google Form CSV
- Automatic footprint calculation for all students
- Dashboard with:
    - KPIs (average score, total users, highest impact)
    - Score distribution charts
    - Demographic comparisons
    - Sustainability leaderboard
    - Campus-level recommendations

5. SYSTEM ARCHITECTURE



6. SCORING MODEL

The ecological footprint is computed as:

TotalScore = Transport + Energy + Food + Plastic

- Higher score → Higher environmental impact
- Categories are weighted based on sustainability impact

7. IMPLEMENTATION

- Python 3 - Core development
- Streamlit - Interactive UI & dashboard
- Pandas - Data processing & aggregation
- Plotly - Interactive visualizations
- Custom CSS - Modern UI (violet, black, white theme)

Key Features:

- Automatic CSV processing
- Real-time score calculation
- Dynamic charts & insights
- Recommendation engine

8. OUTCOME

- Improved Awareness: Students understand their impact clearly
- Behavior Change: Personalized tips guide sustainable habits
- Data Insights: Institutions gain actionable environmental data
- Gamification: Leaderboard encourages participation
- Practical Tool: Effective for demos and real deployment

9. CONCLUSION

The Gryfindors Eco-Footprint Dashboard bridges the gap between awareness and action by transforming lifestyle data into meaningful insights.

It empowers:

- Students - to adopt sustainable habits
- Institutions - to implement data-driven policies

This project demonstrates how technology can drive real-world environmental impact and build a sustainable future.